

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 5 (AT2)

Visualization Development Exercise

Question 1 – Chart Selection (Topic 1: Directory of Visualizations)

Task:

For each of the following scenarios, identify the single best chart type from the given visualization directory (Amounts / Distribution / Proportion / x-y Relationship / Geospatial / Network / Temporal / Multivariate) and justify your selection in 1–2 clear sentences.

1. Comparing the population of 8 countries in the current year

Chart Type: Bar Chart (Amounts)

Justification:

A bar chart is the most appropriate visualization for comparing numerical values across multiple categories. Since each country represents a separate category, the height (or length) of the bars clearly highlights differences in population, making it easy to compare countries and identify the highest and lowest populations.

2. Showing how delivery times vary across 5 courier companies

Chart Type: Box Plot (Distribution)

Justification:

A box plot is best suited for displaying the distribution of delivery times for each courier company. It summarizes the median, quartiles, overall spread, and potential outliers, enabling an effective comparison of consistency and variability among the companies.

3. Showing the proportion of a household budget spent on rent, food, transport, and savings

Chart Type: Pie Chart (Proportion)

Justification:

A pie chart is ideal for illustrating how individual categories contribute to a whole. It clearly

represents the percentage share of each household expense, allowing viewers to quickly understand which category consumes the largest or smallest portion of the total budget.

4. Showing the relationship between hours studied and exam marks for 50 students

Chart Type: Scatter Plot (x–y Relationship)

Justification:

A scatter plot is the most suitable chart for analyzing the relationship between two continuous numerical variables. It helps identify trends, correlations, clusters, or unusual observations, making it easier to determine whether increased study hours are associated with higher exam scores.

5. Showing average temperature for each month across 3 different cities at once

Chart Type: Line Chart (Temporal)

Justification:

A line chart is the best choice for displaying data that changes over time. Using separate lines for each city allows viewers to compare monthly temperature trends, observe seasonal variations, and identify similarities or differences in temperature patterns across the three cities.

6. Showing which airports are directly connected by flight routes

Chart Type: Network Diagram (Network)

Justification:

A network diagram is specifically designed to represent relationships and connections between entities. In this case, airports are represented as nodes and direct flight routes as connecting links, making it easy to visualize the connectivity and route structure within the flight network.

Question 2 – Bar Plot (Topic 2: Amounts)

Task

Create a horizontal bar plot of Downloads_Millions by App_Category, sorted in ascending order. Use a single-hue color gradient so that the bar color intensity reflects the download count. Add the exact download value as a text label at the end of each bar.

R Program

```
library(ggplot2)

# Create data frame
df <- data.frame(
  App_Category = c("Social Media",
                    "Gaming",
                    "Productivity",
                    "Finance",
                    "Health & Fitness",
                    "Education"),
  Downloads_Millions = c(58,46,27,19,33,24)
)

# Sort in ascending order
df <- df[order(df$Downloads_Millions), ]

# Create horizontal bar plot
ggplot(df,
  aes(x = reorder(App_Category, Downloads_Millions),
      y = Downloads_Millions,
      fill = Downloads_Millions)) +

  geom_col(width = 0.7) +

  coord_flip() +

  geom_text(aes(label = Downloads_Millions),
    hjust = -0.2,
    size = 4) +
```

```

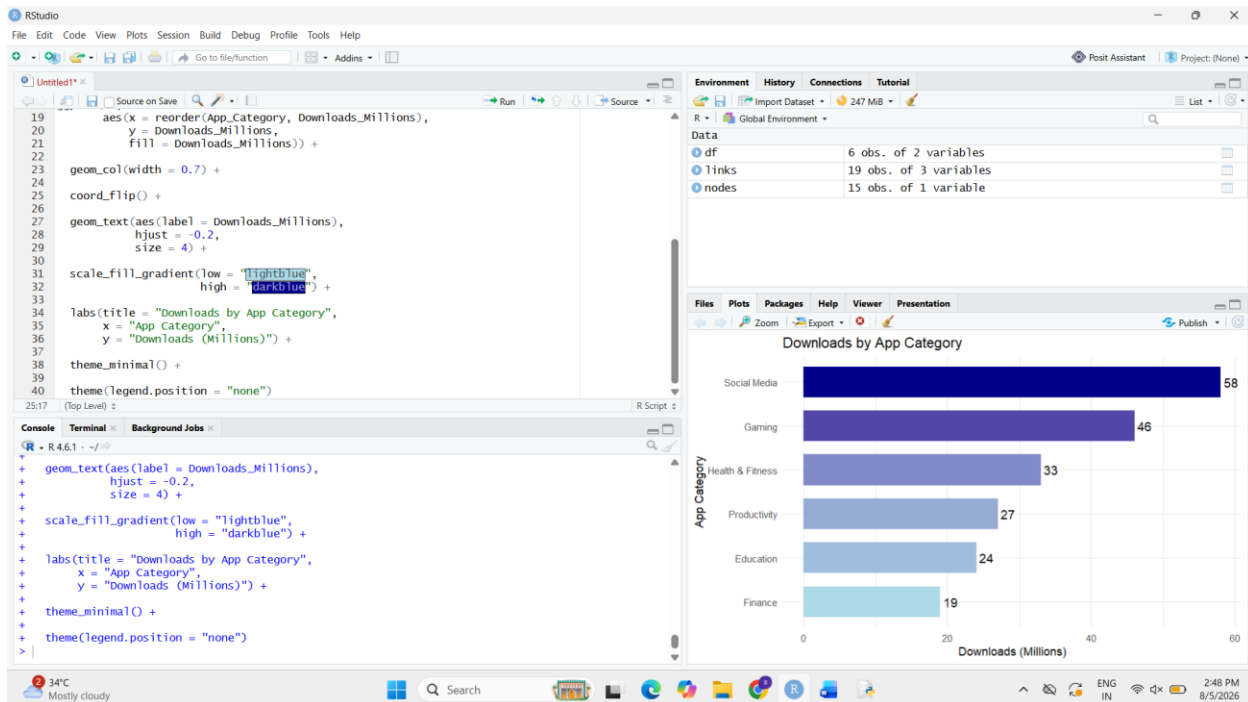
scale_fill_gradient(low = "lightblue",
                    high = "darkblue") +

labs(title = "Downloads by App Category",
      x = "App Category",
      y = "Downloads (Millions)") +

theme_minimal() +

theme(legend.position = "none")

```



Interpretation

The horizontal bar plot presents the number of downloads across different application categories in millions. The categories are arranged in ascending order, making it easy to compare their popularity. Finance has the lowest download count with 19 million, followed by Education (24 million) and Productivity (27 million). Health & Fitness records 33 million downloads, while Social Media has the highest count at 58 million.

Gaming has 46 million downloads. Social Media has the highest download count at 58 million, indicating that it is the most widely downloaded application category among the given categories.

Justification

A horizontal bar plot is the most suitable visualization for comparing numerical values across different categories because it provides a clear comparison of download counts. Sorting the bars in ascending order improves readability and helps viewers quickly identify the categories with the lowest and highest downloads. The single-hue color gradient visually represents the magnitude of downloads, while the value labels display the exact counts, making the chart both informative and easy to interpret.

Question 3 – Grouped/Stacked Bars, Dot Plot & Heatmap (Topic 2: Amounts)

Task

Using the given website_traffic.csv dataset, create the following visualizations:

- Grouped bar chart showing Visits_000s by Month, grouped by Device.
- Stacked bar chart showing total monthly traffic with Device stacked within each month.
- Dot plot showing the total 4-month traffic per device, ranked from highest to lowest.
- Heatmap with Device as rows and Month as columns, color-coded by Visits_000s.

Grouped Bar Chart

R Program

```
library(ggplot2)
```

```
# Create data frame
```

```
df <- data.frame(
```

```
  Device = c(rep("Desktop", 4),
```

```
             rep("Mobile", 4),
```

```
             rep("Tablet", 4)),
```

```
  Month = rep(c("Jan", "Feb", "Mar", "Apr"), 3),
```

```
  Visits_000s = c(45, 48, 42, 50,
```

```
                 80, 88, 95, 102,
```

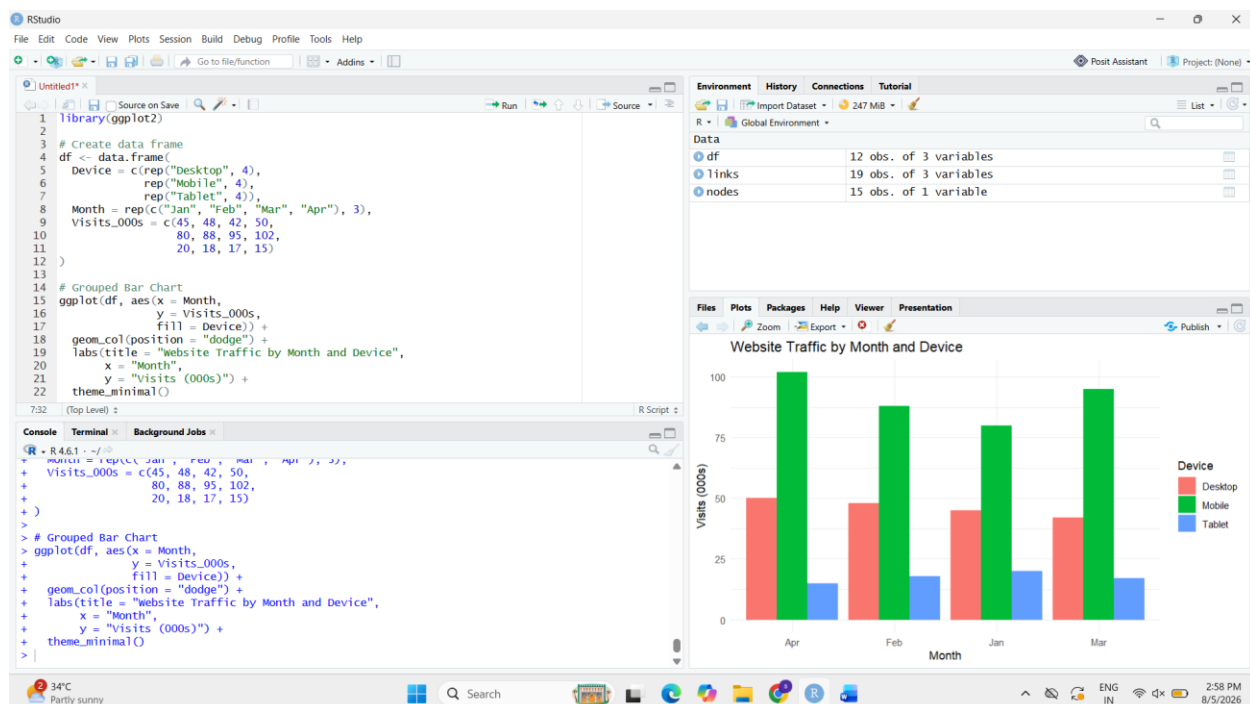
```

20, 18, 17, 15)

)

# Grouped Bar Chart
ggplot(df, aes(x = Month,
               y = Visits_000s,
               fill = Device)) +
  geom_col(position = "dodge") +
  labs(title = "Website Traffic by Month and Device",
       x = "Month",
       y = "Visits (000s)") +
  theme_minimal()

```



Interpretation

The grouped bar chart compares website traffic generated by Desktop, Mobile, and Tablet devices for each month. Mobile records the highest number of visits in every month, Desktop shows moderate traffic, while Tablet consistently has the lowest traffic.

Justification

A grouped bar chart is suitable because it enables direct comparison of different devices within the same month, making differences in website traffic easy to identify.

Stacked Bar Chart

R Program

```
library(ggplot2)

# Create data frame
df <- data.frame(
  Device = c(rep("Desktop", 4),
             rep("Mobile", 4),
             rep("Tablet", 4)),
  Month = rep(c("Jan", "Feb", "Mar", "Apr"), 3),
  Visits_000s = c(45, 48, 42, 50,
                 80, 88, 95, 102,
                 20, 18, 17, 15)
)

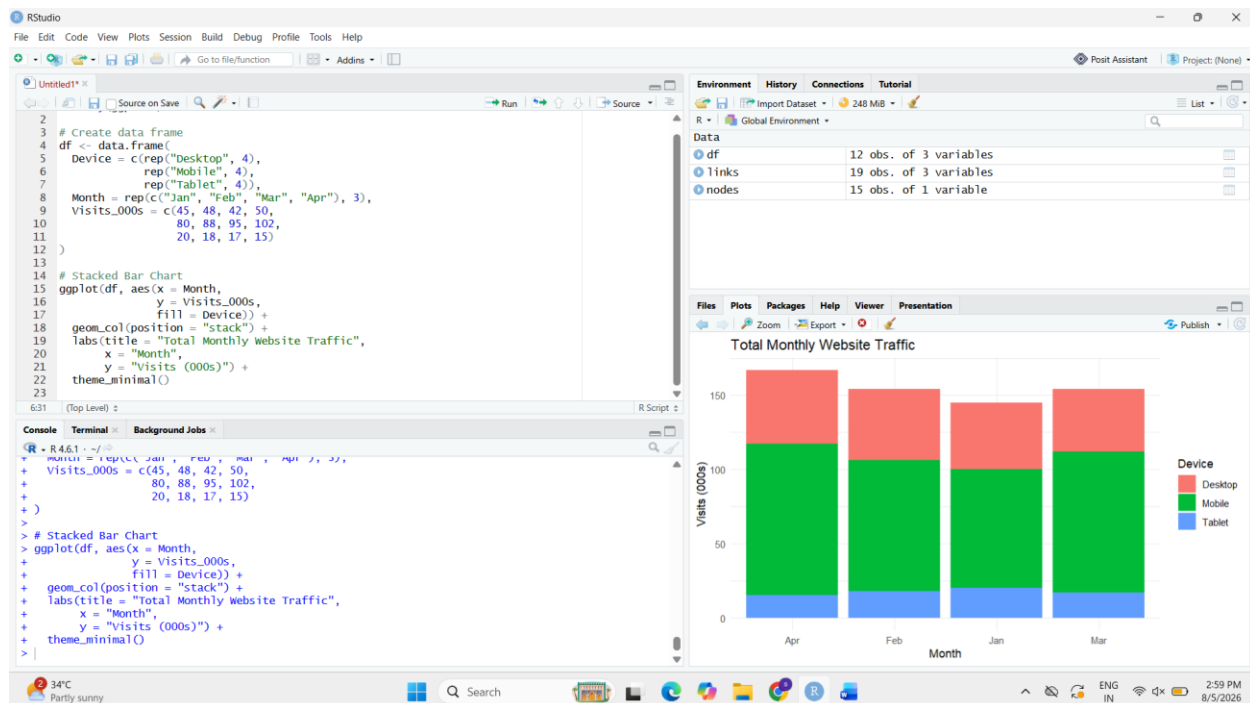
# Stacked Bar Chart
ggplot(df, aes(x = Month,
               y = Visits_000s,
               fill = Device)) +
  geom_col(position = "stack") +
```

```
labs(title = "Total Monthly Website Traffic",
```

```
  x = "Month",
```

```
  y = "Visits (000s)") +
```

```
theme_minimal()
```



Interpretation

The stacked bar chart displays the total website traffic for each month while also showing the contribution of each device. Total traffic increases from January to April, mainly due to the continuous increase in Mobile traffic.

Justification

A stacked bar chart is appropriate because it simultaneously displays the overall monthly traffic and the contribution of each device, providing a clear understanding of cumulative values.

Dot Plot

R Program

```
library(ggplot2)
```



```

# Create data frame
df <- data.frame(
  Device = c(rep("Desktop", 4),
             rep("Mobile", 4),
             rep("Tablet", 4)),
  Month = rep(c("Jan", "Feb", "Mar", "Apr"), 3),
  Visits_000s = c(45, 48, 42, 50,
                 80, 88, 95, 102,
                 20, 18, 17, 15)
)

# Calculate total visits
total <- aggregate(Visits_000s ~ Device, data = df, sum)

# Rank from highest to lowest
total <- total[order(-total$Visits_000s), ]

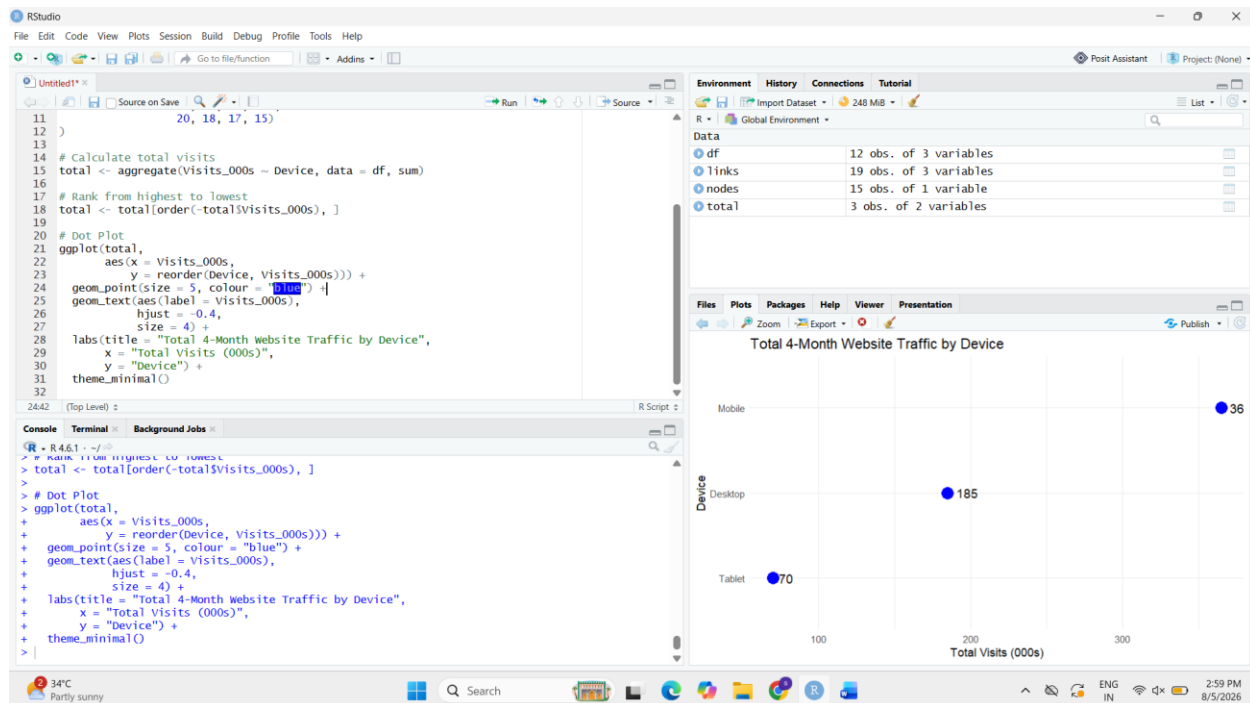
# Dot Plot
ggplot(total,
  aes(x = Visits_000s,
      y = reorder(Device, Visits_000s))) +
  geom_point(size = 5, colour = "blue") +
  geom_text(aes(label = Visits_000s),
    hjust = -0.4,
    size = 4) +
  labs(title = "Total 4-Month Website Traffic by Device",

```

```

x = "Total Visits (000s)",
y = "Device") +
theme_minimal()

```



Interpretation

The dot plot ranks devices based on their total website traffic over four months. Mobile has the highest traffic with **365 thousand visits**, Desktop records **185 thousand visits**, and Tablet has the lowest traffic with **70 thousand visits**.

Justification

A dot plot is an effective visualization for displaying ranked values. It presents individual totals clearly while minimizing visual clutter, making comparisons between devices straightforward.

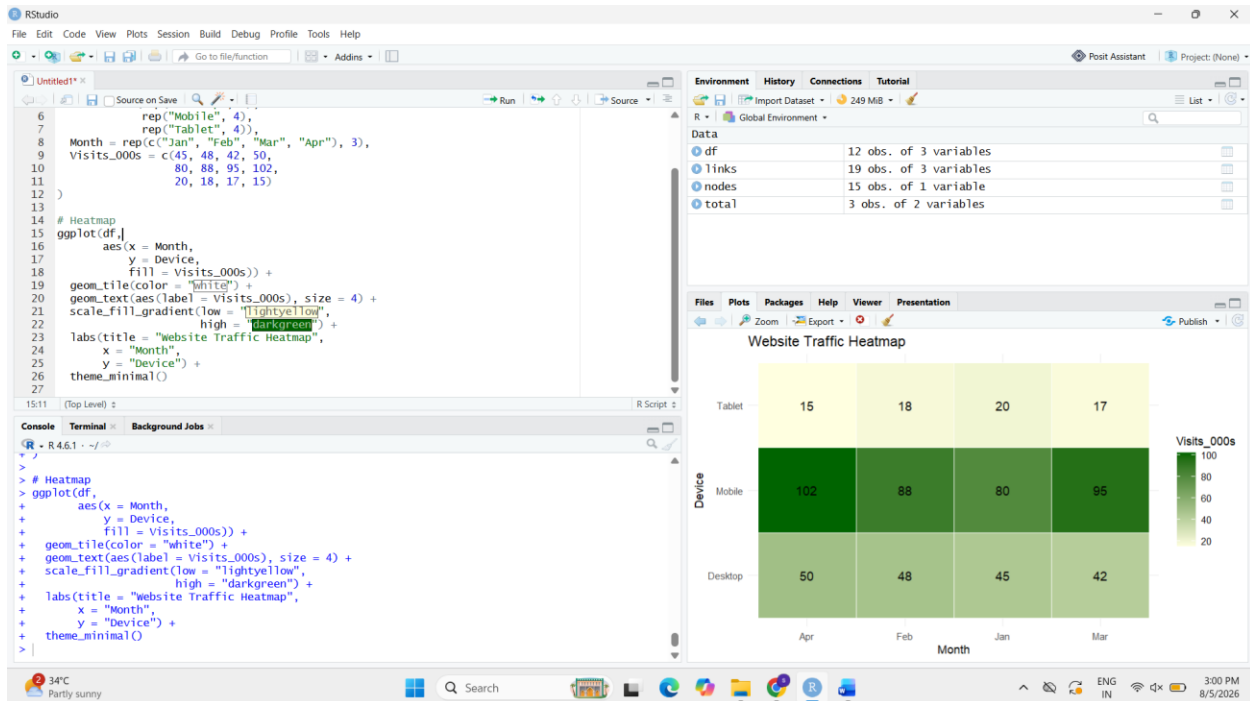
Heatmap

R Program

```
library(ggplot2)
```

```
# Create data frame
df <- data.frame(
  Device = c(rep("Desktop", 4),
             rep("Mobile", 4),
             rep("Tablet", 4)),
  Month = rep(c("Jan", "Feb", "Mar", "Apr"), 3),
  Visits_000s = c(45, 48, 42, 50,
                 80, 88, 95, 102,
                 20, 18, 17, 15)
)

# Heatmap
ggplot(df,
  aes(x = Month,
      y = Device,
      fill = Visits_000s)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Visits_000s), size = 4) +
  scale_fill_gradient(low = "lightyellow",
                     high = "darkgreen") +
  labs(title = "Website Traffic Heatmap",
       x = "Month",
       y = "Device") +
  theme_minimal()
```



Interpretation

The heatmap uses color intensity to represent website traffic across different devices and months. Darker shades indicate higher traffic, showing that Mobile consistently receives the highest number of visits, particularly in April, while Tablet has the lowest traffic throughout the observed period.

Justification

A heatmap is appropriate because it efficiently compares values across two categorical variables using color intensity. It helps viewers quickly identify trends, patterns, and areas of high and low website traffic.

Question 4 – Violin & Ridgeline Plots (Topic 3: Distributions)

Task

Generate a synthetic dataset using the following distributions with a **random seed of 7**:

- Route A: 50 delivery times $\sim$ Normal(mean = 32, sd = 6)
- Route B: 50 delivery times $\sim$ Normal(mean = 28, sd = 4)
- Route C: 50 delivery times $\sim$ Normal(mean = 40, sd = 9)
- Route D: 50 delivery times $\sim$ Normal(mean = 35, sd = 5)

Create:

- A **violin plot** with an embedded boxplot.
- A **ridgeline plot** showing overlapping density curves ordered by mean delivery time (fastest at the bottom).

Violin Plot

R Program

```
library(ggplot2)
```

```
# Set random seed
```

```
set.seed(7)
```

```
# Generate synthetic data
```

```
Route <- rep(c("Route A", "Route B", "Route C", "Route D"), each = 50)
```

```
Delivery_Time <- c(
```

```
  rnorm(50, mean = 32, sd = 6),
```

```
  rnorm(50, mean = 28, sd = 4),
```

```
  rnorm(50, mean = 40, sd = 9),
```

```
  rnorm(50, mean = 35, sd = 5)
```

```
)
```

```
df <- data.frame(Route, Delivery_Time)
```

```
# Violin Plot
```

```
ggplot(df, aes(x = Route, y = Delivery_Time, fill = Route)) +
```

```
  geom_violin(trim = FALSE) +
```

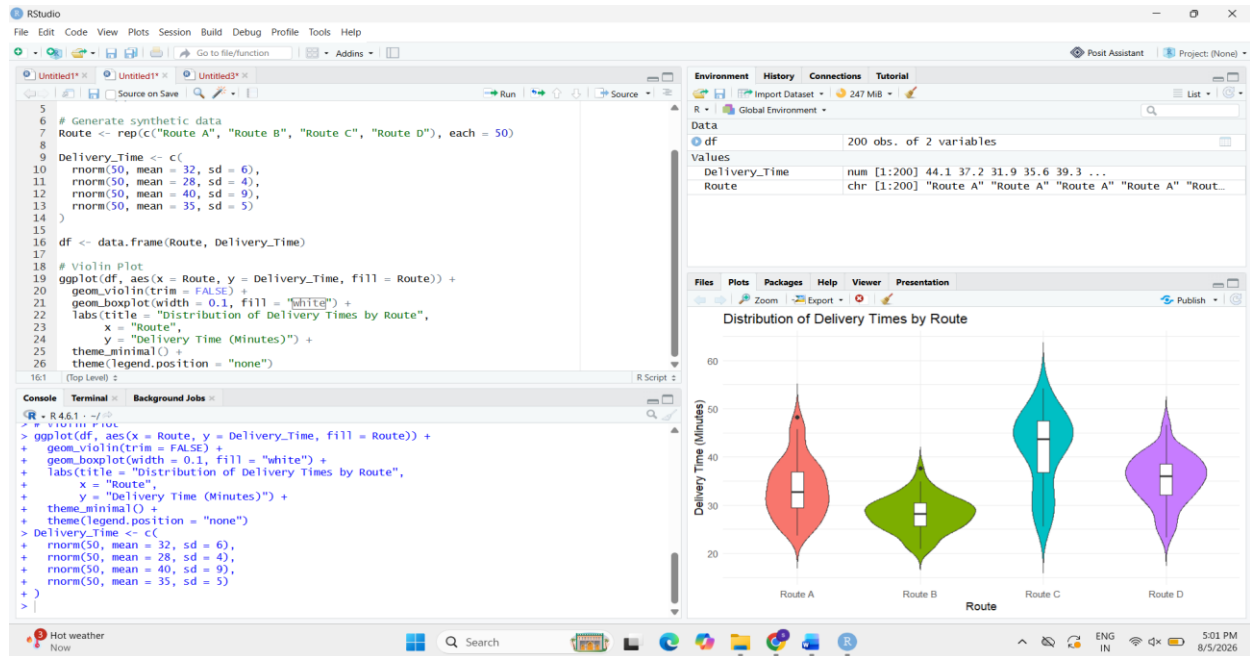
```
  geom_boxplot(width = 0.1, fill = "white") +
```

```
  labs(title = "Distribution of Delivery Times by Route",
```

```

x = "Route",
y = "Delivery Time (Minutes)" +
theme_minimal() +
theme(legend.position = "none")

```



Interpretation

The violin plot shows the distribution of delivery times for all four routes. Route B has the lowest average delivery time, indicating the fastest route, while Route C has the highest average delivery time and the widest spread, showing greater variability. Routes A and D exhibit moderate delivery times with relatively balanced distributions.

Justification

A violin plot is suitable because it combines the features of a boxplot and a density plot. It provides information about the median, spread, and overall distribution shape, making it effective for comparing delivery times across multiple routes.

Ridgeline Plot

R Program

```

library(ggplot2)
library(ggribbles)

```

```
# Set random seed
set.seed(7)

# Generate synthetic data
Route <- rep(c("Route A", "Route B", "Route C", "Route D"), each = 50)

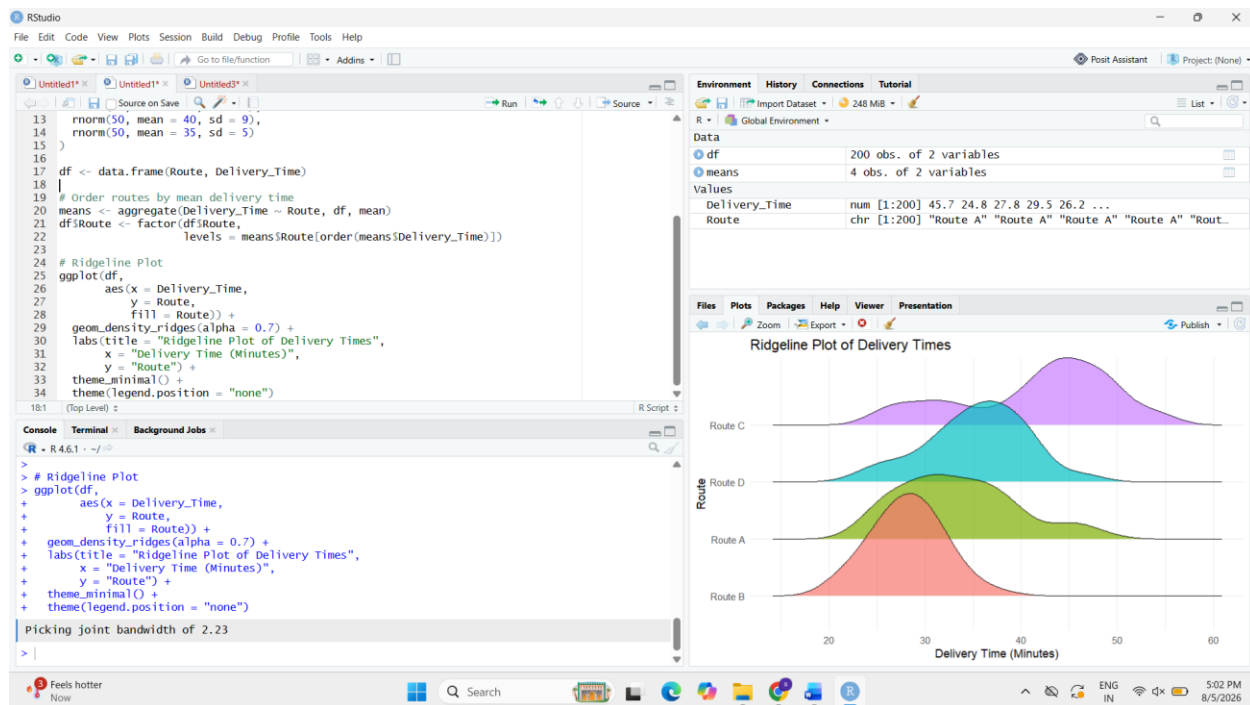
Delivery_Time <- c(
  rnorm(50, mean = 32, sd = 6),
  rnorm(50, mean = 28, sd = 4),
  rnorm(50, mean = 40, sd = 9),
  rnorm(50, mean = 35, sd = 5)
)

df <- data.frame(Route, Delivery_Time)

# Order routes by mean delivery time
means <- aggregate(Delivery_Time ~ Route, df, mean)
df$Route <- factor(df$Route,
  levels = means$Route[order(means$Delivery_Time)])

# Ridgeline Plot
ggplot(df,
  aes(x = Delivery_Time,
    y = Route,
    fill = Route)) +
  geom_density_ridges(alpha = 0.7) +
```

```
labs(title = "Ridgeline Plot of Delivery Times",
      x = "Delivery Time (Minutes)",
      y = "Route") +
theme_minimal() +
theme(legend.position = "none")
```



Interpretation

The ridgeline plot compares the density distributions of delivery times for all routes. Route B appears at the bottom because it has the lowest average delivery time, while Route C appears at the top with the highest average delivery time and the greatest variation. The overlapping density curves make it easy to compare the distribution patterns among the routes.

Justification

A ridgeline plot is appropriate because it visualizes multiple distributions in a compact and organized manner. It highlights differences in the shape, spread, and central tendency of each route, making comparisons more effective than viewing separate density plots.

Question 5 – Histograms & Density Plots (Topic 3: Distributions)

Task

Use the built-in iris dataset and the Petal.Length variable to:

- Plot a histogram with 20 bins.
- Overlay a kernel density estimate (KDE) curve on the same plot.
- Repeat the histogram and density plot by Species (Setosa, Versicolor, Virginica) to compare the distributions.

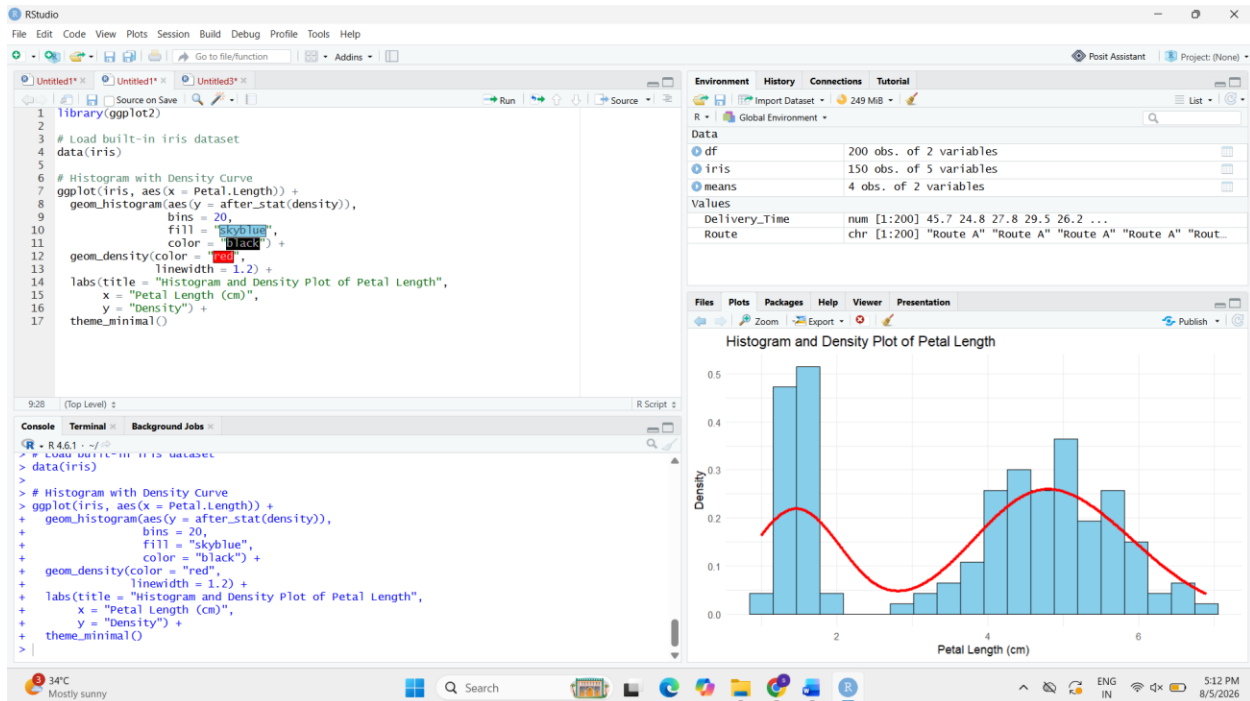
Histogram with Density Curve

R Program

```
library(ggplot2)

# Load built-in iris dataset
data(iris)

# Histogram with Density Curve
ggplot(iris, aes(x = Petal.Length)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 20,
    fill = "skyblue",
    color = "black") +
  geom_density(color = "red",
    linewidth = 1.2) +
  labs(title = "Histogram and Density Plot of Petal Length",
    x = "Petal Length (cm)",
    y = "Density") +
  theme_minimal()
```



Interpretation

The histogram displays the distribution of petal lengths in the iris dataset, while the density curve provides a smooth representation of the distribution. The data are not perfectly symmetric and show multiple peaks, indicating that the petal lengths vary across different iris species.

Justification

A histogram is appropriate for displaying the frequency distribution of a continuous variable, while the density curve provides a smooth estimate of the underlying distribution. Combining both visualizations offers a clearer understanding of the data's shape and spread.

Histogram with Density Curve by Species

R Program

```
library(ggplot2)
```

```
# Load built-in iris dataset
```

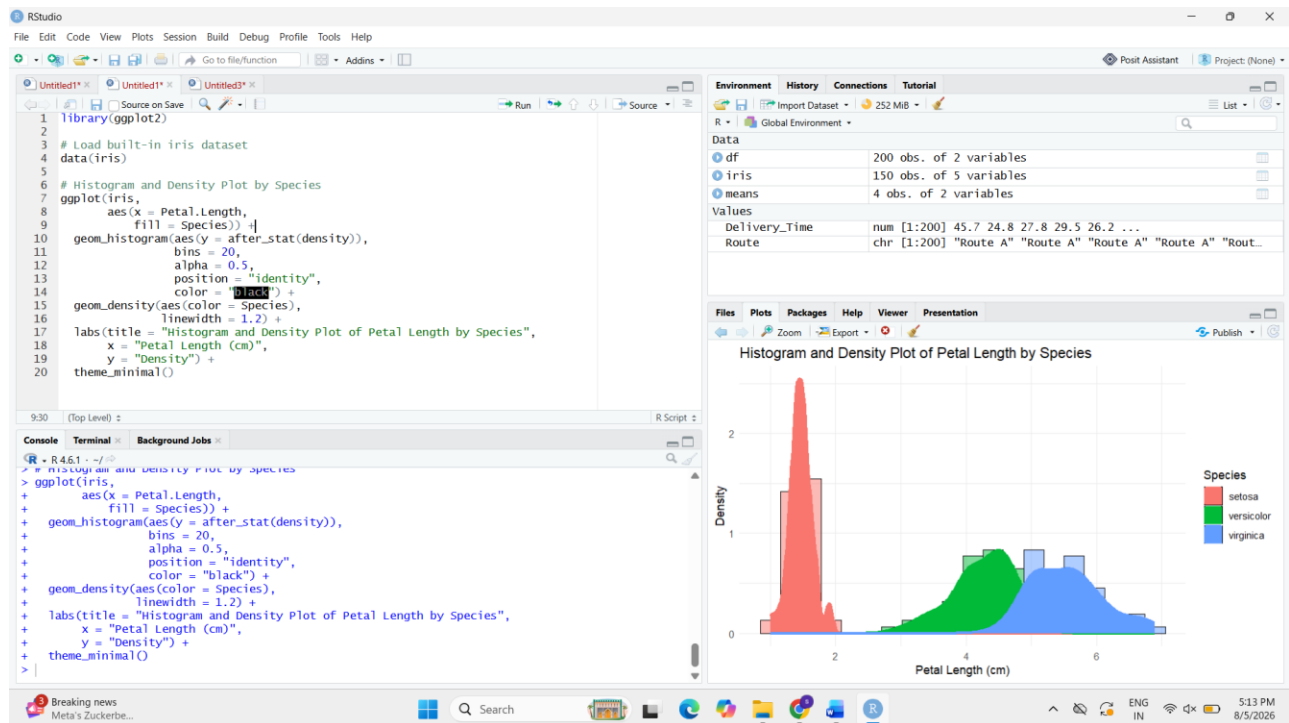
```
data(iris)
```

```
# Histogram and Density Plot by Species
```

```

ggplot(iris,
      aes(x = Petal.Length,
          fill = Species)) +
  geom_histogram(aes(y = after_stat(density)),
                bins = 20,
                alpha = 0.5,
                position = "identity",
                color = "black") +
  geom_density(aes(color = Species),
              linewidth = 1.2) +
  labs(title = "Histogram and Density Plot of Petal Length by Species",
       x = "Petal Length (cm)",
       y = "Density") +
  theme_minimal()

```



Interpretation

The grouped histogram and density curves compare the distribution of petal lengths among the three iris species. **Setosa** has the shortest petal lengths with a narrow distribution, **Versicolor** shows intermediate petal lengths, and **Virginica** has the longest petal lengths with a relatively wider distribution. The three species form distinct distribution patterns, making their differences easy to observe.

Justification

Using different colors for each species enables direct comparison of their petal length distributions. Overlaying density curves on the histograms highlights differences in shape, central tendency, and variability, making this visualization effective for comparing multiple groups.

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